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Plasmon-driven high harmonic generation of benzene: effect of spatial inhomogenity of near field

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Abstract

When short and intense laser pulses interact with nanosystems which include plasmonic nanoparticles (NPs) and molecules, the laser induced spatially inhomogeneous near-fields modify the molecule's electron dynamics. Consequently, this affects the high-order harmonic generation (HHG) processes. Although a lot of experimental and theoretical work have been devoted to study the plasmon-driven HHG processes of atoms or diatomic molecules, there are no clear pictures to show how the electronic trajectories and spectral characters vary with the locations of polyatomic molecules. Here we perform a comprehensive theoretical study on the coupled dynamics between the spatially inhomogeneous fields and matter by combining the computational electrodynamic model with the time-dependent density functional theory. The plasmon-driven electron dynamics and HHG spectra of a benzene molecule in the vicinity of a gold nanosphere or in the gap of a pair of nanospheres have been investigated. The unique phenomena arisen from the surface-enhanced near-field are described and the underlying mechanism which determines the salient features of plasmon-driven HHG is presented. Our study demonstrates that the HHG spectral features are sensitive to the molecule-NP's distance and the incident light's polarization, and that the intensities of lower-order harmonics follow an exponentially decay with the molecule-NP's distance and the laser's polarization direction. Additionally, we find that when the incident light frequency is resonant with plasmon excitation, the obvious phase difference between the x- and y-components of near fields makes the electronic trajectories obviously follow elliptical paths. However, that elliptically polarized light does not generate the strikingly different spectra due to the near field's spatial inhomogeneity.

1. Introduction

When atomic or molecular gas sample is illuminated by an intense femtosecond laser, the light with frequencies much greater than the original will be emitted. This nonlinear conversion process is known as HHG [1, 2], which is a well-known method of producing coherent extreme ultraviolet light. However, HHG process requires high pulse intensities, greater than 10^{13} W cm^{-2}, which are not directly attainable only using the output power of a femtosecond oscillator. Chirped-pulse amplification is required [3].

Plasmonic metallic nanoparticles (MNPs) are characterized by their strong interaction with resonant photons through an excitation of surface plasmon resonance (SPR). SPR origins from the resonant photon-induced collective oscillation of conduction electrons and it can be characterized by a build-up of intense, spatially inhomogeneous oscillating electric fields nearby the metallic nanostructure [4, 5]. These fields have the highest intensity at the surface of the nanostructure and decrease exponentially with distance from the surface, which can modify the laser-driven electron dynamics of molecules in vicinity of nanostructures. Consequently, this has important impact on pivotal processes such as above-threshold ionization and high-order harmonic generation [6]. For example, in 2008, Kim *et al* [7] demonstrated a method of HHG that required no extra cavities by exploit-

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ing the local field enhancement induced by resonant plasmons within a metallic nanostructure consisting of bow-tie-shaped gold elements on a sapphire substrate. In their experiment, a laser pulse of low intensity (about 10^{11} W cm^{-2}) was directly focused onto a bow-tie nanoantenna array with the injection of xenon gas, ending up with the produce of XUV wavelengths from the 7th (114 nm) to the 21st (38 nm) harmonics.

HHG features are determined by the sample's geometric and electronic structures and the spatial distribution of applied electric field as well. When the applied field is homogeneous, the selection rule (SR) for HHG spectra can be deduced on the basis of point-dipole approximation. For example, in linearly polarized (LP) fields, only odd harmonics are observed for the centrosymmetric molecules [8, 9], while in circularly polarized (CP) lights, only $kN \pm 1$ order harmonics ($k = 0, 1, 2...$) are present for target molecule with $N-$ fold rotational symmetry [9–11]. The spatial symmetry of the HHG process can be broken either by the use of an asymmetric driving laser field or an asymmetric generation medium. As a result, other photon frequencies rather than the SR-allowed harmonics will be observed. Since the near field generated by MNPs has large field gradient, this spatial inhomogeneity caused by the strong local confinement definitely breaks the SR and leads to the plasmon-driven HHG spectra differing significantly with those only induced by the femtosecond laser, which has been proofed by many experiments [12–14] and theoretical investigations [15–18].

The measurement and theoretical simulation of the HHG emission make the possibility to extract useful microscopic dynamical and structural information of the sample. The elegant and simple three-step quasi-static model [19, 20], (1) tunnel ionization by an intense low-frequency laser field, (2) acceleration of the free electron, and (3) recollision with nuclei, has been widely-used to interpret atomic HHG. Due to the multiple nuclei and their relative motions of polyatomic molecule [21], the measurement and simulation of molecular HHG spectra are far more difficult. Besides, molecular harmonics are very sensitive to the molecular orientation, conformations, and dynamic motion of the electrons [22, 23]. The experiments on HHG of polyatomic molecules are thus far sparse. A few of theoretical works have devoted to investigate the mechanism of HHG of polyatomic molecules using a variety of theoretical approaches [10, 11, 24–28], and revealed quite different mechanics than the simple three-step quasi-static model. When the sample is set close to a plasmonic nanostructure, the coupled dynamics between the MNP-induced spatially inhomogeneous fields and matter should be taken into account, which further complicates the electronic dynamics. Thus recent investigations on plasmon-driven molucular HHG were only limited on the simplest diatomic systems like H_2^+, where the model of one electron in an effective potential was applied [29–32].

To simulate the optical properties of hybrid structures composed of NPs and molecules or semiconductors, the most accurate and frequently adopted methods are the mixed quantum-classical methods, where the light absorption/scattering by plasmonic element is obtained via classical electromagnetic approaches such as the finite difference time domain (FDTD) technique, while the optical responses of the molecule are evaluated quantum mechanically. For example, Schatz's [33, 34], Chen's [35], Neuhauser's [36] and Jensen's [37] groups applied the mixed quantum/classical approaches which combine TDDFT or TDHF for adsorbates with the classical electrodynamics methods for the large metal NPs to investigate the surface-enhanced optical properties or the electron transport in nanodevices. By adopting TDDFT/FDTD approach, our group investigated the plasmon enhanced light absorption and revealed the dependency of plasmonic enhancement of molecular absorption with the energy gap between molecular excitation and plasmon mode [38]. In those previous works, the molecule was supposed to be small enough so that its shape is neglected, therefore the field applied to the molecule was treated uniformly and the dipole approximation was used. In fact, due to the strong local confinement, the surface near field is strongly spatially nonuniform. For a polyatomic molecule immersed in this nonuniform electric field, its optical responses are determined not only by the dipole moment but also by the higher multipole moments (permanent as well is induced) [39]. It is noted that some previous works did account for the effect of field's inhomogeneity by fitting the near field as a function of spatial coordinate in linear [40], polynomial [41] or exponential [18, 42] forms, then applied this space-resolution field to the molecules. Yamaguchi *et al* [43] theoretically studied the SR of plasmon-enhanced second harmonic generation (SHG) of a symmetric and an asymmetric molecules by identifying the source of optical near field as an oscillating electric dipole.

The goal of this paper is directed to an understanding of the coupled dynamics between these spatially inhomogeneous fields and matter, and the effect of proximity of metal nanostructures on the polyatomic HHG spectra. We aim to make a quantitative description of the changes of spectral characters caused by the interaction between the molecule and plasmonic metal NPs and depict the quantitative dependency of surface near-fields and the optical signature enhancement ratio on the geometrical parameters of superstructures, such as the NP-molecule separation distance, number of NPs, and the molecular locations as well. Furthermore, we will demonstrate how the phase differences between the incident laser field and near field of NPs, and between the field components along and perpendicular to the polarization direction affects the electronic dynamics of the molecule. The mixed TDDFT/FDTD scheme is applied to simulate the plasmon-driven HHG of benzene molecule. The hybrid systems are constructed by setting a benzene molecule near a plasmonic Au nanosphere or in the gap of a pair of Au nanospheres. We firstly obtain the time-dependent spatially nonuniform near field generated by

NPs by FDTD technique, then describe the molecular electronic evolution by the time-dependent Kohn–Sham equation without imposing the perturbation expansion. With this mixed quantum-classical scheme, the molecular dynamic responses to the two driving fields, the strong surface scattered field and the incident laser field, can be simultaneously described without the requirement of empirical parameters. The field-matter interaction and the molecular internal Coulomb forces have also been treated on equal footing. The mixed TDDFT/FDTD scheme was used to calculate the plasmon enhanced absorption spectra and surface-enhanced Raman spectra [38, 44]. Here we extend this mixed scheme to include the effect of applied electric field's non-uniformity.

The rest of the paper is arranged as follows. In section 2, we briefly describe the mixed TDDFT/FDTD scheme. The computational details of real-time real-space TDDFT and the FDTD method are shown. In section 3, we first investigate the dependence of surface scattered field on the numbers of NPs, the surface separation distance of NP pairs, and the molecular locations. Then, we demonstrate the calculated surface-enhanced HHG spectra of benzene molecule near Au NPs. Finally, the concluding remark is given in section 4.

2. Theory and computational details

HHG can be viewed as the power radiated by a system in the strong field, which depends on the acceleration of electrons [45],

$$\ddot{d}(t) = \frac{\mathrm{d}^2\langle\Psi(\mathbf{r},t)|\hat{\mu}|\Psi(\mathbf{r},t)\rangle}{\mathrm{d}t^2}, \tag{1}$$

where $\hat{\mu}$ is the molecular dipolar operator and $\Psi(r,t)$ is the time-dependent wave function which can be obtained by solving the time-dependent Schrödinger equation. Alternatively, the dipole acceleration can be written as

$$\ddot{d}(t) = \frac{\mathrm{d}^2[\int \mathrm{d}\mathbf{r}\hat{\mu}\rho(\mathbf{r},t)]}{\mathrm{d}t^2}, \tag{2}$$

with respect to the time-dependent electron density $\rho(\mathbf{r},t)$, which can be obtained by solving the time-dependent Kohn–Sham (KS) equation,

$$\begin{aligned} \mathrm{i}\frac{\partial\varphi_p(\mathbf{r},t)}{\partial t} =& \{-\frac{1}{2}\nabla^2 + v_{\mathrm{ion}}(\mathbf{r},t) + v_H[\rho_t](\mathbf{r},t) \\ &+ v_{xc}[\rho_t](\mathbf{r},t) + v_{\mathrm{field}}(\mathbf{r},t)\}\varphi_p(\mathbf{r},t), \end{aligned} \tag{3}$$

where φ_p stands for the pth KS orbital. The adiabatic local density approximation coupled with an average-density self interaction correction (ADSIC) [46] is adopted in this work. The motivation to choose ADSIC is because of its correct asymptotic behaviour for a large distance, $v_{xc} \propto -1/r$. $v_{\mathrm{field}}(t)$ denotes the matter-field interaction. For the isolated molecule, the applied field is laser field $\mathbf{F}_0(t)$, while for the nanostructure constructed by the NPs and the molecule, the electric field applied to molecule is the near field, $\mathbf{F}(\mathbf{r},t) = \mathbf{F}_0(t) + \mathbf{F}_{\mathrm{sca}}(\mathbf{r},t)$. $\mathbf{F}_{\mathrm{sca}}(\mathbf{r},t)$ is the scattered field generated by Au NPs, which is obtained by solving the Maxwell's equations using the FDTD technique within the FDTD + + package [47], here we adopt a simulation box of $60 \times 60 \times 60\,\mathrm{nm}^3$ for the single NP and a box of $80 \times 60 \times 60\,\mathrm{nm}^3$ for a pair of NPs. The grid size is set to be 0.167 nm, and the time step is 3.05×10^{-4} fs. The dielectric function of gold is described by one Drude model term (conduction band electron motion), adding five Lorentz oscillators (interband electron transitions) [48] as following form to well fit the experimental dielectric function [49],

$$\epsilon(\omega) = \epsilon_\infty - \frac{{\omega_D}^2}{\omega(\omega + \mathrm{i}\gamma_D)} - \sum_{n=1}^{5}\frac{\Delta\epsilon_{Ln}{\omega_{Ln}}^2}{\omega(\omega + 2\mathrm{i}\delta_{Ln}) - {\omega_{Ln}}^2}. \tag{4}$$

The detailed fitting parameters are shown in table 1. The comparison with the experimental curvature can be found in figure S1 in supporting information (SI) (stacks.iop.org/EST/1/044001/mmedia).

In our calculations, we assume the incident light is a x-polarized laser pulse with a finite duration: $\mathbf{F}_0(t) = A\sin^2(\frac{2\omega_0}{n}t)\cos(\omega_0 t)\hat{\mathbf{e}}_{\mathbf{x}}, (0 \leqslant t \leqslant \frac{n\pi}{\omega_0})$, where incident field amplitude $A = 0.274\,\mathrm{V\,\mathring{A}^{-1}}$ (corresponding intensity maximum $I_0 = 10^{12}$ W cm^{-2}). The optical cycle n is set to be 20, and two different incident wavelengths of $\lambda_0 = 800$ and 496 nm are chosen, which correspond to the wavelength of common used Ti-sapphire laser and the plasmonic resonance frequency of the gold nanosphere, respectively.

The open-source TDDFT code in Octopus software [50, 51] is employed to solve the time-dependent KS equation in real time and real space. We employ a spacing of $\Delta x = 0.22$ Å and a spherical simulation box with a radius of $R_m = 20$ Å for the real space grid. The complex absorbing potential [52] with a form of $V_{\mathrm{CAP}}(r) = -\mathrm{i}\eta\sin^2[(r - R_m + L)\pi/2L]$ for $R_m - L \leqslant r \leqslant R_m$; and $V_{\mathrm{CAP}}(r) = 0$ for elsewhere, is adopted to smoothly deplete the wave function so that we can eliminate reflection at the grid boundary. L is the absorbing width which is set to be 5 Å in this work. To make sure the settings are reasonable, we check the result convergence of the HHG spectra with respect to Δx, R_m and L, and demonstrate the results in figure S2 in SI. It is clear that the

Table 1. Fitting parameters in dielectric function of gold.

	Drude		Lorentz		
ε_∞	ω_D (eV)	γ_D (eV)	$\Delta\varepsilon_L$	ω_L (eV)	δ_L (eV)
			0.40	2.76	0.30
			0.50	3.15	0.40
3.00	8.8	0.072	1.84	4.15	1.00
			1.60	6.00	2.00
			−0.50	1.84	0.42

parameters we adopt in the calculations can produce the accurate results. Since the grid spacing for FDTD and TDDFT is different, in this work we use the modified quadratic Shepard algorithm [53] to interpolate the FDTD field into a finer mesh at every time step (an illustration of the compatibility between FDTD and TDDFT grids can be found in figure S3 in SI). The explicit time-propagation is performed using the approximated enforced time-reversal symmetry method [50], and the 4th order Taylor expansion method is used to the approximate the exponential of the Hamiltonian. The time step is set to be 1.22×10^{-3} fs which is four times of the FDTD time step, so that every instant field needed in TDDFT calculations can be extracted from FDTD steps, and the stability is tested for all systems.

3. Results and discussion

3.1. The near-field generated by NPs

At first, we calculate the absorption spectra of a single gold nanosphere with the different radius and a pair of nanospheres with different spatial separation values of g. The results are shown in figure S4 in SI. The absorption of single NP clearly shows a plasmon resonance peak at 2.5 eV. This plasmon peak of NP dimer shows a red-shift and an increase of absorption intensity with the decreasing spatial separation. Since the polarization is along dimer axis, only the longitudinal modes are excited. The peak's red-shift can be explained by using the plasmon hybridization model [54]. The absorption spectrum of isolated benzene molecule is also calculated and shown in figure S5 in SI. Its excitation energies are much larger than that of plasmon resonance's dipole mode.

The surface near-fields generated by single NP and the NP dimer are presented in figure 1, which shows clearly that the strong near-field appears close to the surface, and the so-called 'hot spots' in the gap of NP dimer are formed. In figure 1, we mark several selected molecular locations: $X_1(12,0), X_2(15,0), X_3(18,0)$ around the single NP and $Z_1(-1.5,0), Z_2(-1,0), Z_3(-0.5,0), Z_0(0,0)$ in the gap of NP dimer. The output fields at those points under incident lights of different wavelengths are shown in figure 2. The x-components of the fields are significant and the field intensity decreases as the order of $X_1 > X_2 > X_3$, indicating that there is a notable field gradient along the polarization direction. However, it is noted that the field in dimer gap is almost uniform. For small NPs, the output field seems just simple scaling up of the incident field: the maximum amplification can be 5 times, which leads to the intensity of 2.5×10^{13} W cm^{-2} (the corresponding field intensity for each molecular location is shown in table S1 of SI). When the NP's plasmon excitation is resonant with the incident light of 496 nm, the near field shows a time lag due to the polarization effect and the field keeps oscillating in a small amplitude even when the driving field fades away.

Besides the points along the x-axis, we follow a circular path along the single NP sphere to choose other molecular locations: X_1, U, W, V, and Y, which have the same distances of 12 nm from the NP center, and corresponding angles $\theta = 0°, 30°, 45°, 60°, 90°$ (see figure 1). The time-dependent output fields at U, W, V are shown on the middle panels of figure 2, and they are more complicated because both x and y components of the field are obvious now. At resonance condition, where the wavelength of incident light is equal to 496 nm ($\omega_0 = 2.5$ eV), there is an obvious phase difference between the x and y components of the output field for the points of out of x-axis.

The phase differences between the output field and incident electric field, and between the x and y components of the output field for the points out of x-axis can be explained by the analytical expression of the output field of a single nanosphere in the electrostatic approximation [55],

$$\mathbf{F}_{\text{total}} = \mathbf{F}_0 + \mathbf{F}_{\text{sca}} = \mathbf{F}_0 + \frac{3\hat{\mathbf{r}}(\hat{\mathbf{r}}\cdot\mathbf{p}) - \mathbf{p}}{4\pi\epsilon_0 r^3},$$

$$\mathbf{p} = 4\pi\epsilon_0 R^3 \frac{\epsilon - 1}{\epsilon + 2}\mathbf{F}_0,$$

where ϵ_0 is the vacuum permittivity, r is the distance from origin, $\hat{\mathbf{r}}$ is the unit vector in the direction of the point of interest, and $\mathbf{p}$ is the field-induced polarization in NP. Obviously, both F_x and F_y vary with θ, and $F_y = 0$ at $\theta = 0°$ or 90° (x- or y-axis). Because the electrons oscillate in response to the applied electromagnetic field, and

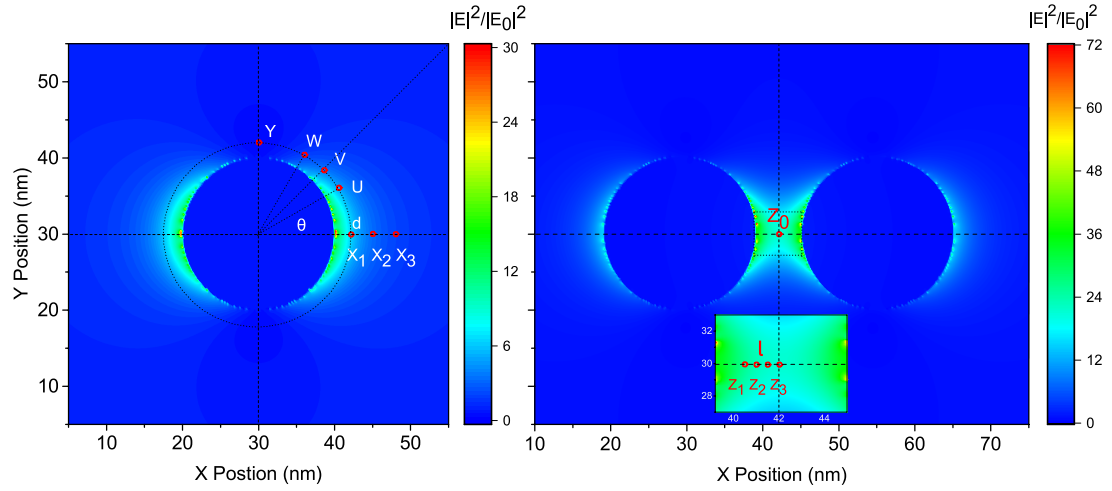


Figure 1. The contour plot ($z = 0$ plane) of the electric field enhancement $|E(\omega)|^2/|E_0(\omega)|^2$ (ω corresponding to the absorption peak at 496 nm) of a NP monomer and dimer with $R = 10$ nm and $g = 6$ nm.

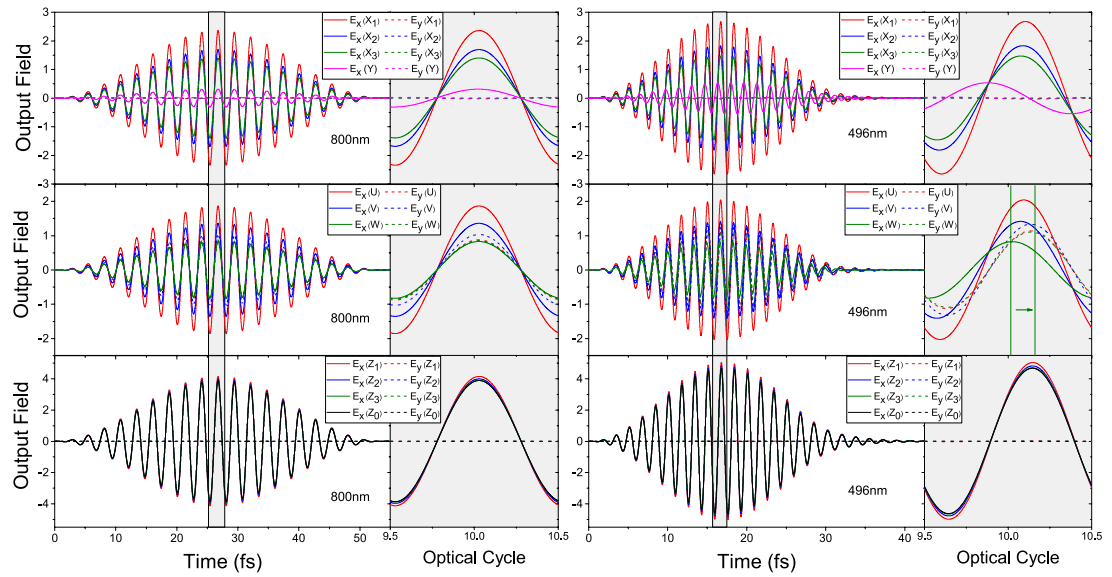


Figure 2. The output electric fields at selected locations around NP structure. Both x and y components are shown for the points of X_1, X_2, X_3, Y (upper panel), U, W, V (middle panel), and Z_1, Z_2, Z_3, Z_0 (lower panel).

their motion is damped via collision occurring with a characteristic frequency γ_D, a phase difference between the macroscopic polarization $\mathbf{p}$ and $\mathbf{F_0}$ is thus induced. The value of phase difference is sensitive to the incident light frequency and γ_D. From figure 2, we observe that at $\lambda_0 = 496$ nm, there are largest phase delays between $\mathbf{p}$ and $\mathbf{F_0}$, and between the x and y components of the output field for the points of U, W, and V. The obvious phase difference existed in the x and y components of the output field will induce the electron circular motion, and the generation of elliptical high harmonics.

3.2. Generation of elliptical harmonics

Here we pay our attention to the situations where molecule is set at off-x-axis locations. We plot the variations of μ_x and μ_y with the simulating time for benzene in figure 3, in order to show the averaged trajectory of electron motion on the xy-plane during the molecular excitations. At $\lambda_0 = 800$ nm, the electrons are just oscillating along the direction of applied field. They are gradually tilted away from x axis as θ increases (however, the tilt angle is not equal to θ). At $\lambda_0 = 496$ nm, the electronic trajectories obviously follow elliptical pathes when the molecule is set at the points of U, V, and W, among which the W situation has the highest eccentricity.

The circular motion of electrons can also be directly observed in real time. The snapshots of induced electron density at some selected instants (when the field reaches its maximum) are displayed in figure 4. At $\lambda_0 = 800$ nm, the averaged trajectory is in a line, and the electrons of benzene move mainly through C–H and C–C bonds. At $\lambda_0 = 496$ nm, the averaged trajectory is elliptical, and the electrons may hop between neighboring H atoms

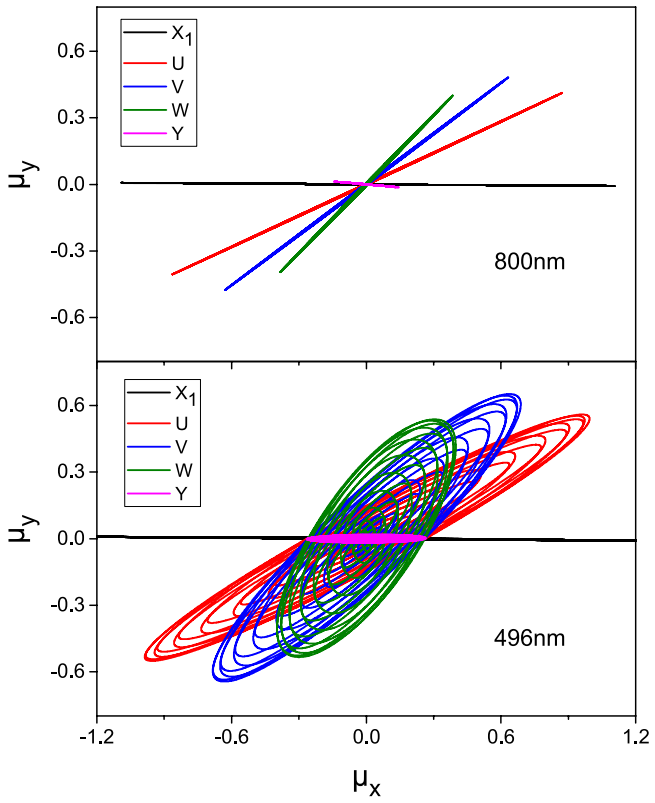


Figure 3. The variations of induced dipolar moment components, μ_x and μ_y, during the simulation time.

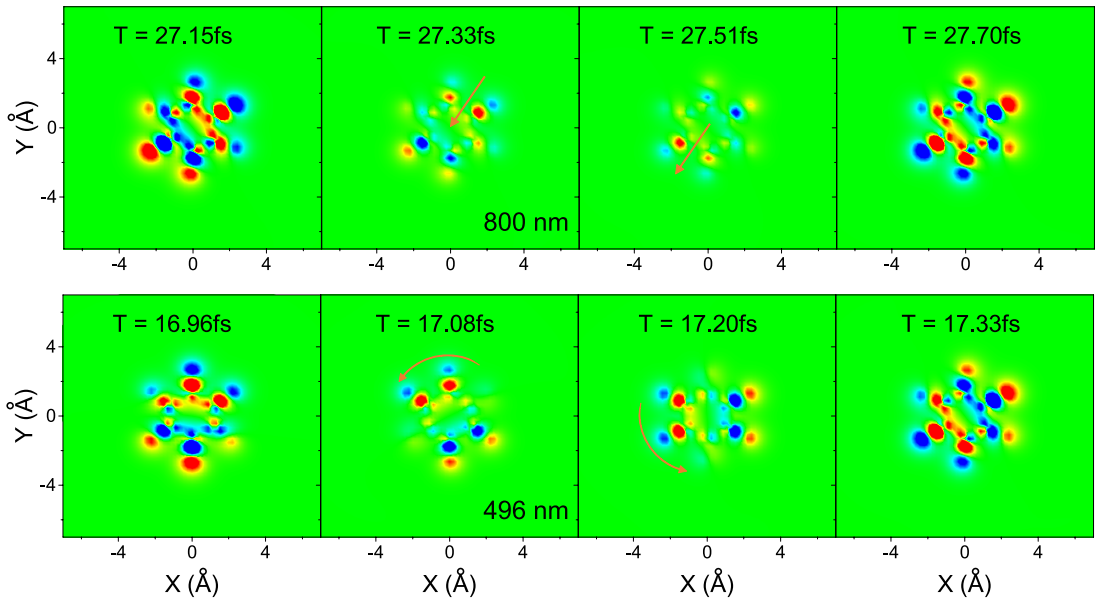


Figure 4. The snapshots of induced electron densities (isovalue = 0.001) at $\lambda_0 = 800$ and 496 nm. Four time points are selected, and the molecular center of mass (MCM) is set at point V ($\theta = 30°$).

besides move through bonds. The electron dynamics of benzene located in point U, V, W situations are made as movies and can be found in SI (movies.mp4). If we watch the movies of electron motion, we can see that the electrons are actually spinning around each atoms.

This kind of electron motion is attributed to the elliptical electric forces, which can be deduced from the previous analysis of the output fields. Figure 5 shows the calculated ellipticity, [56] defined as $e = |\frac{r^2+1-\sqrt{(r^2+1)^2-4r^2\sin^2\phi}}{2r\sin\phi}|$, where $r = |A_x|/|A_y|$ denotes the amplitude ratio of dipole accelerations and $\sin\phi = \mathrm{Im}\{A_x/A_y\}/r$ denotes the phase difference of the two components. $e = 0$ and $e = 1$ mean that the generated harmonics are linearly polarized and circularly polarized, respectively. Since $A_{x,y}$ is the fourier transform of dipole acceleration $A(t)$, the

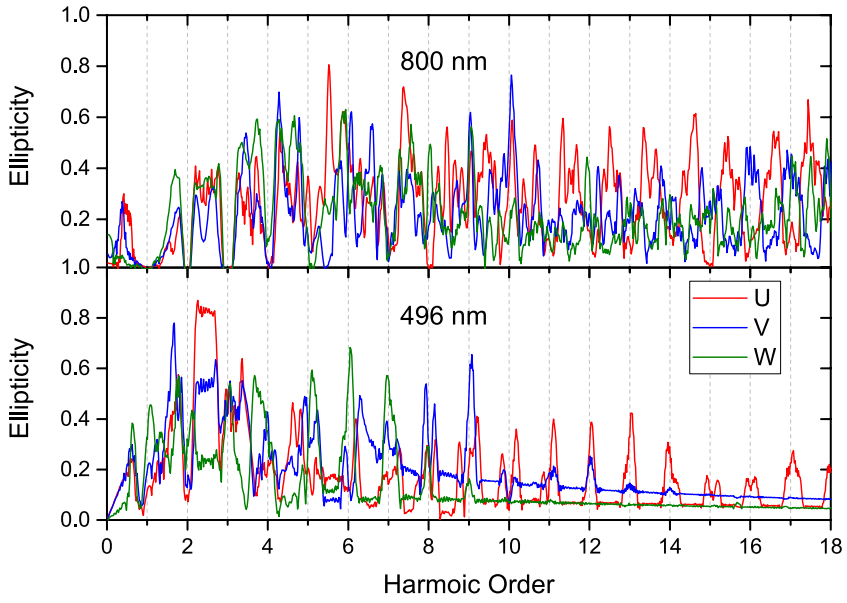


Figure 5. The ellipticity of high harmonics. The MCM is set at U, V and W, respectively.

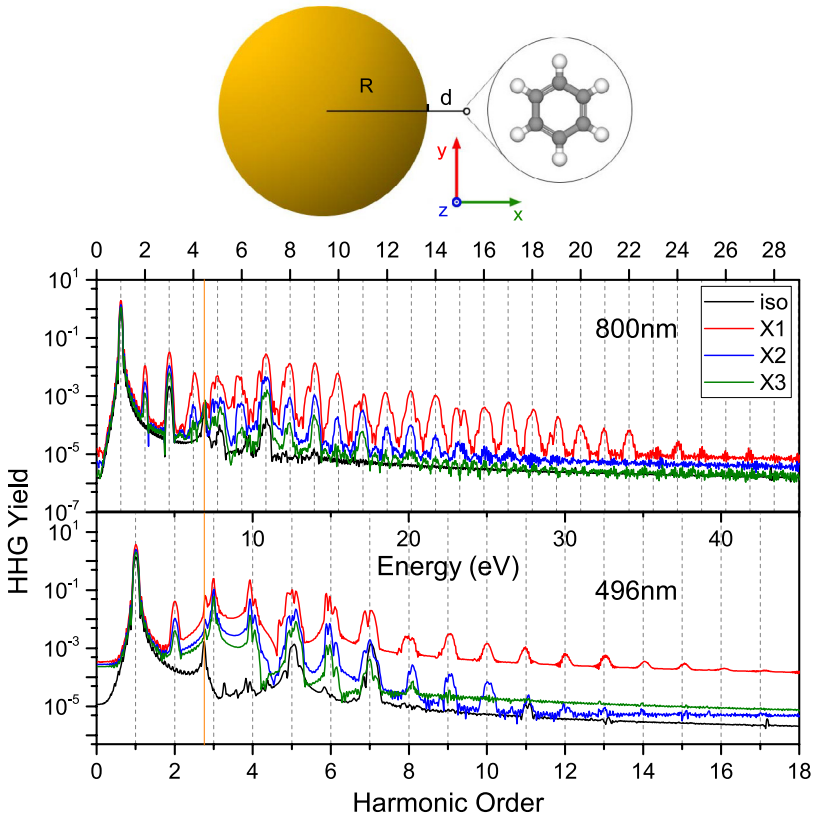


Figure 6. The plasmon-driven HHG spectra of benzene close to a Au nanosphere with $R = 10$ nm at incident wavelengths of 800 and 496 nm. The MCM is set at X_1, X_2 and X_3, respectively.

ellipticity is a function of harmonic order. For example, the value e of the first order harmonic stands for the ellipticity of the driving field. In figure 5, we show how the ellipticity varies with the harmonic order for the benzene set around Au NP sphere. At $\lambda_0 = 800$ nm, e shows noise pattern and has no obvious peak in integer orders. At $\lambda_0 = 496$ nm, the driving field in the location of W undoubtedly has the largest ellipticity of $e = 0.4$ as shown in figure 3), and the corresponding values are 0.1 and 0.2 for U and V locations, respectively. In this case, there are obvious peaks in high integer orders, and some ellipticity of high order harmonics are larger than that of the first order. This could be a promising way to generate circularly polarized high harmonics [57].

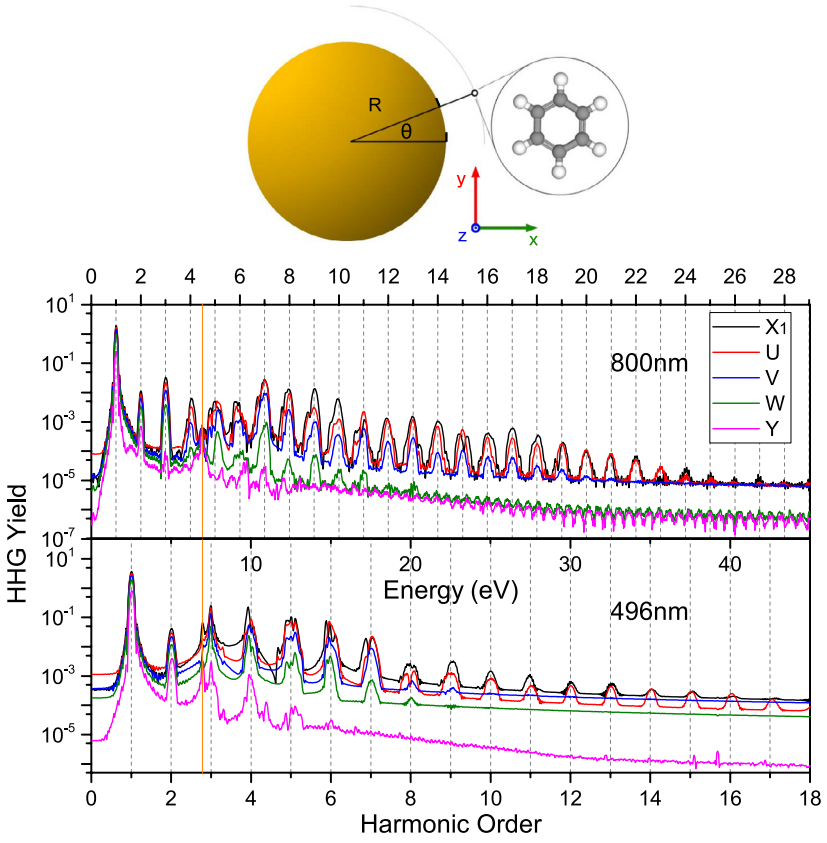


Figure 7. The plasmon-driven HHG spectra of benzene close to a Au nanosphere with $R = 10$ nm at incident wavelengths of 800 and 496 nm. The MCM is set at X_1, U, V, W and Y, respectively.

3.3. Plasmon-driven HHG spectra of benzene at different position

We calculate the plasmon-driven HHG spectra of the centro-symmetric benzene molecule, which will be set close to an Au nanosphere or in the gap of nanosphere dimer. Figure 6 shows the molecular HHG spectra with different distances between benzene and nanosphere. The HHG spectra of isolated molecule under the incident laser fields are shown for comparison. For the isolated molecule, the incident electric field can only generate low-order harmonics. We can easily find a non-integer peak appeared at 7.0 eV with both incident wavelengths, which has the same energy with the molecular electronic transition (the location of first absorption peak in figure S5). If we look closer, other resonance peaks corresponding to molecular excited states can be found in higher energy [58]. In fact, the broad absorption band ranged from 10 to 25 eV extends the HHG cutoff to 13th and 7th order, while the predicted cutoff position is below 10 eV.

When the molecule is set close to a NP, its high harmonics become more and more clear as the distance d reduces. Typical HHG spectrum shows a structure of 'descent-plateau-cutoff', with a cutoff position formula: $E_{\text{cutoff}} = I_p + 3.17U_p$, where I_p and $U_p \propto |\varepsilon|^2\lambda^2$ are the ionization potential and the ponderomotive energy. In our simulation, though the ponderomotive energy of the plasmonic output field is smaller than the ionization energy, the maximum harmonic can reach to 40 eV. The simple three step model which tackles one electron moving with one nucleus in uniform field, thus cannot be used to explain the plasmon enhanced polyatomic molecular HHG. Comparing the two incident wavelengths, we observe that the cutoff position at 800 nm is higher than at 496 nm. However, larger wavelength of the incident laser field does not always have higher cutoff energy.

In figure 6, we can also see that the HHG spctra of isolated molecule has only odd orders, but both even and odd orders are shown when the NP presents. To have a clear physical insight on the origin of even harmonics, we calculate the HHG spectra of benzene subjected to the inhomogeneous (real near-field) and homogeneous (assumed that the field applied to the molecule is same with that at the MCM) fields, respectively (see figure S6 in SI). The results show that in low energy part the two-types of fields yield the odd harmonics with nearly same amplitudes, and even harmonics only appear when the molecule interacts with the nonuniform near fields. It is clear that the large field gradient existed in the near-field results in the appearance of even harmonics. As the benzene molecule is set away from NP surface ($X_1 \rightarrow X_3$), the discrepancy between two results becomes not obvious.

In SI, we give an explanation why the even harmonics appear when the applied field is non-uniform. The nth even harmonics appear only when

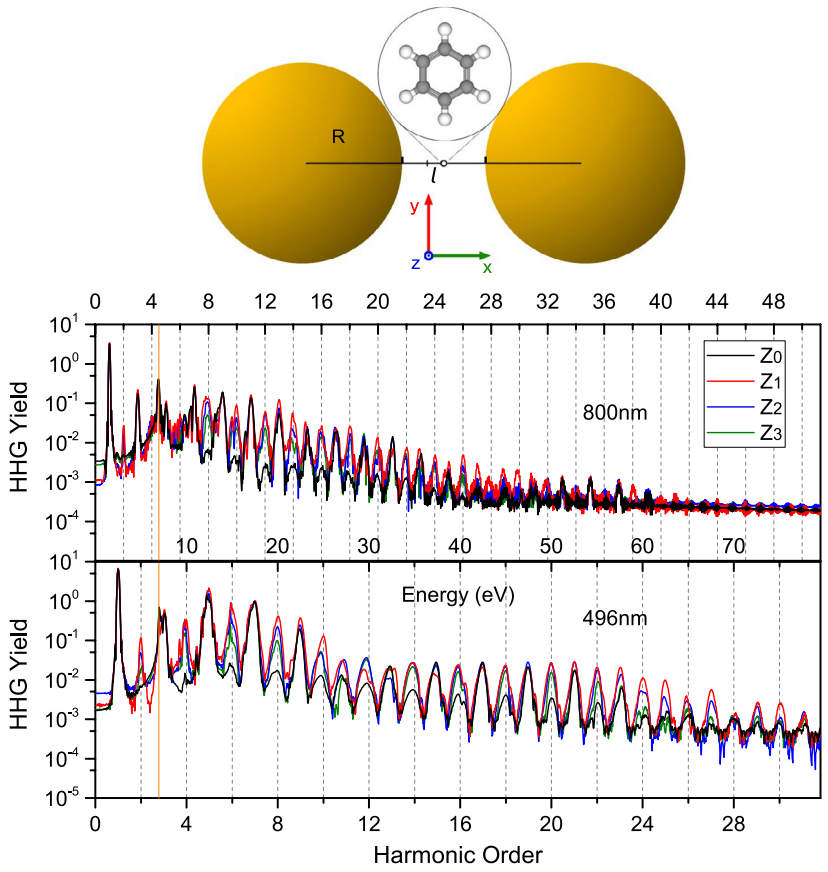


Figure 8. The plasmon-driven HHG spectra of benzene in the gap of NP dimer with $R = 10$ nm and $g = 6$ nm (at the points of Z_1, Z_2, Z_3 and Z_0).

$$\langle 0|\mu|k\rangle \underbrace{\langle k|V_0|l\rangle\langle l|\cdots|p\rangle\langle p|V_0|0\rangle}_{n} \neq 0. \tag{5}$$

It is hardly satisfied when the field is uniform, where $\hat{V}_0 = -\mu \cdot F_0$. In nonuniform field, the mater-field interaction can be expanded as $\hat{V} = -\sum_q \mu_q F_{0,q} - \frac{1}{3}\sum_{qq'} \Theta_{qq'} F_{qq'} + \cdots$. In this case the multipole-field interaction terms will play a role so that the above condition for the transition matrix elements may be satisfied when the applied field is inhomogeneous.

Figure 7 shows the calculated plasmon-driven HHG spectra of benzene molecule located at the points of X_1, U, W, V and Y around $R = 10$ nm Au NP sphere. Here the distance between the NP center and MCM is kept to be 12 nm, and the angles between the light polarization direction and the unit vectors of these positions vary from $\theta = 0 \rightarrow 90°$. As θ increases, the molecule moves away from the hot spot, the output near field becomes weaker and weaker, and the HHG intensity reduces and the cutoff position drops to low energy. Since now both x and y components of the near field contribute to the harmonics, and the power spectrum is determined by dipole accelerations in x and y directions: $|\mathbf{A}(\omega)|^2 = A_x^2 + A_y^2$, the HHG spectra of benzene at the location of Y also present even harmonics (caused by the field gradient in y direction), although the fields have inversion symmetry along x axis.

We also calculated plasmon-driven HHG spectra of benzene subject to the near-field generated by NP dimer (see figure 8). When the molecule is set at the center of the gap (point Z_0, black line), the HHG spectra only show odd harmonics because the near field has inversion symmetry distribution. Once the molecule moves away from the center of the gap, the even harmonics appear. As the position shift from the gap center l increases, the extension of HHG cutoff position and the enhancement of spectral intensity are not very significant, since the field gradient in gap is relatively small. We also notice that the cutoff position at 496 nm (~80 eV) is now higher than 800 nm (~60 eV), which can be explained by larger field amplification ratio when NP dimer system resonates with the incident laser (see lower panel of figure 2).

In addition, the plasmon-driven HHG spectra of NP dimer with varied spatial separations g (the molecule is fixed at gap center) have also been calculated. The results are shown in figure S7 in SI. It is clear that the increase of NP size raises the enhancement ratio of output field and favors the HHG process, and the decrease of dimer gap

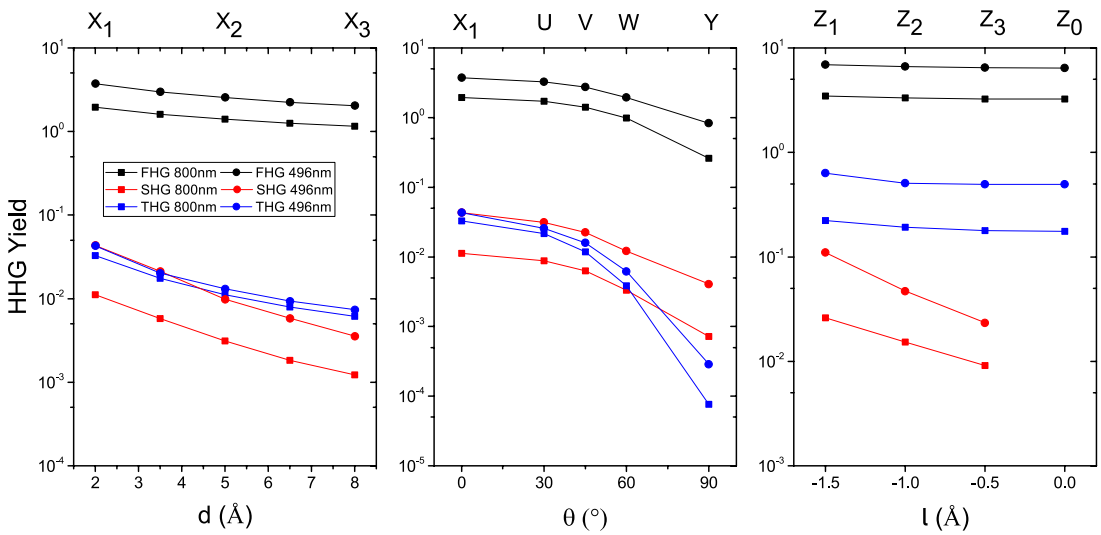


Figure 9. The FHG, SHG and THG intensities vary with the molecular locations. The left and middle panels are for the systems of molecule-NP where the molecular locations varies with d and θ. The right panel is for the system of NP-molecule-NP.

makes a stronger NP coupling, which leads to a more enhanced near field, and the HHG becomes more efficient, with the highest harmonic cutoff energy up to the highest $\sim$100 eV.

To have a clear demonstration on the variance of HHG intensities with the spatial separation between the NPs and molecules, we plot the variation of the first, second and third order harmonics intensities of plasmon-driven HHG with the molecular positions in figure 9. The left panel of figure 9 shows that the FHG, SHG and THG decay exponentially with the increasing separation from NP. The variation trend of SHG agrees with the experimental measurement [59], where an empirical exponentially decay formula was proposed and fitted for the experimental measurement of plasmon-enhanced SHG. The THG intensity shows a quite gentle drop slope, indicating that SHG and THG have different dependency with the fields. The THG intensity is mainly determined by the field intensity, while the SHG is predominantly dependent on the field gradient. A similar pattern can be found in the case of NP dimer (see the right panel of figure 9). With the same position shift from gap center, the change of SHG is larger than that of THG. Note that SHG data for the molecule set in the gap center of the dimer is missing because the even harmonics are forbidden due to the spatial symmetry of the applied field.

4. Concluding remarks

We have systematically investigated the plasmon-enhanced HHG spectra of a benzene molecule in the vicinity of the gold nanosturcture (single NP or NP pair). By tuning the distance between molecule and NP, the spatial separation of a pair of NPs, and the light polarization, one can adjust the output field generated by plasmon excitation and therefore tune the field applied to the molecule. As a result, the molecular HHG spectra are significantly affected. Besides a significantly increased cutoff in plasmon-driven HHG, we observe the even harmonics. Because the spatial resolution of the near field is taken into account in our simulation, by applied near field to molecule, the selection rule for the harmonics is relaxed and even harmonics appear. However, there is an exception when the benzene molecule is put in the gap center of NP dimer, where the inversion symmetry of field still keeps.

The features of harmonics are sensitive to the spatial separation between the molecule and NP, and the field polarization as well. As the spatial separation and the polarization angle increase, SHG and THG intensities decay exponentially.

The wavelength of incident laser affects the spectral shape slightly, and it influences the spectral cutoff. When the incident light frequency is resonant with plasmon excitation, the x and y components of the output near field have obvious phase difference. The electron motion driven by this near field shows elliptical trajectory on the conjugate ring plane for the molecule set at off-x-axis locations. This interesting phenomenon may provide us possibilities in application of elliptical high harmonics.

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Supporting information

Mechanism of even harmonics generation in plasmon-driven HHG; the parameters in dielectric function; compatibility of FDTD and TDDFT grids; absorption spectra of NPs and isolated benzene; comparison of HHG with real near-field and homogeneous field; influence of the NP dimer's spatial separation on HHG; animation of induced electron densities (movies.mp4).

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